

SAE 12 : S'initier aux réseaux informatiques

Rapport de la SAE 13

Société 6 - RT1 - A2

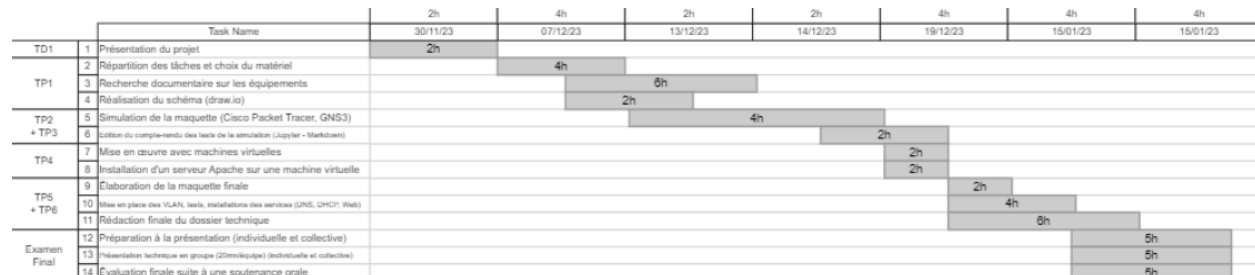


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GANTT

Tout d'abord nous avons réalisé un gant pour nous permettre de nous organiser, lors du déroulement du projet.



Matrice RACI

Nous avons donc répartis les tâches entre nous, sous forme d'une matrice RACI, nous permettant de nous organiser et de nous préparer lors du passage final de cette SAE.

Tâches	Loïc	Baptiste
1. Analyse et Planification	- Analyser le cahier des charges.	- Examiner les ressources en appui.
	- Définir les besoins spécifiques.	- Comprendre les compétences techniques.
	- Élaborer un plan d'implémentation.	- Identifier les éléments du cahier des charges.
2. Configuration du réseau	- Établir un schéma annoté.	- Recenser les services demandés.
	- Établir les caractéristiques des équipements.	- Définir le plan d'adressage IPv4.
	- Éditer le devis du matériel.	- Configurer le serveur web sur le réseau ADMIN.
	- Configurer les VLANs.	
	- Installer le serveur DHCP et DNS.	
3. Maquette virtuelle	- Utiliser "draw.io" pour le schéma.	- Réaliser la simulation avec Cisco Packet Tracer.
	- Documenter le schéma.	- Rédiger le compte-rendu des tests de simulation.
4. Maquette réelle (machines virtuelles)	- Mise en œuvre des machines virtuelles.	- Installation d'un serveur Apache.
	- Configurer les VLANs sur la maquette.	
	- Préparer le dossier technique.	- Tests, capture de trames, installations des services.
	- Élaborer la maquette finale avec les étudiants RT2.	
5. Documentation et Présentation	- Préparer la présentation technique orale.	- Assurer la conformité avec le cahier des charges.
	- Participer à l'évaluation en groupe.	

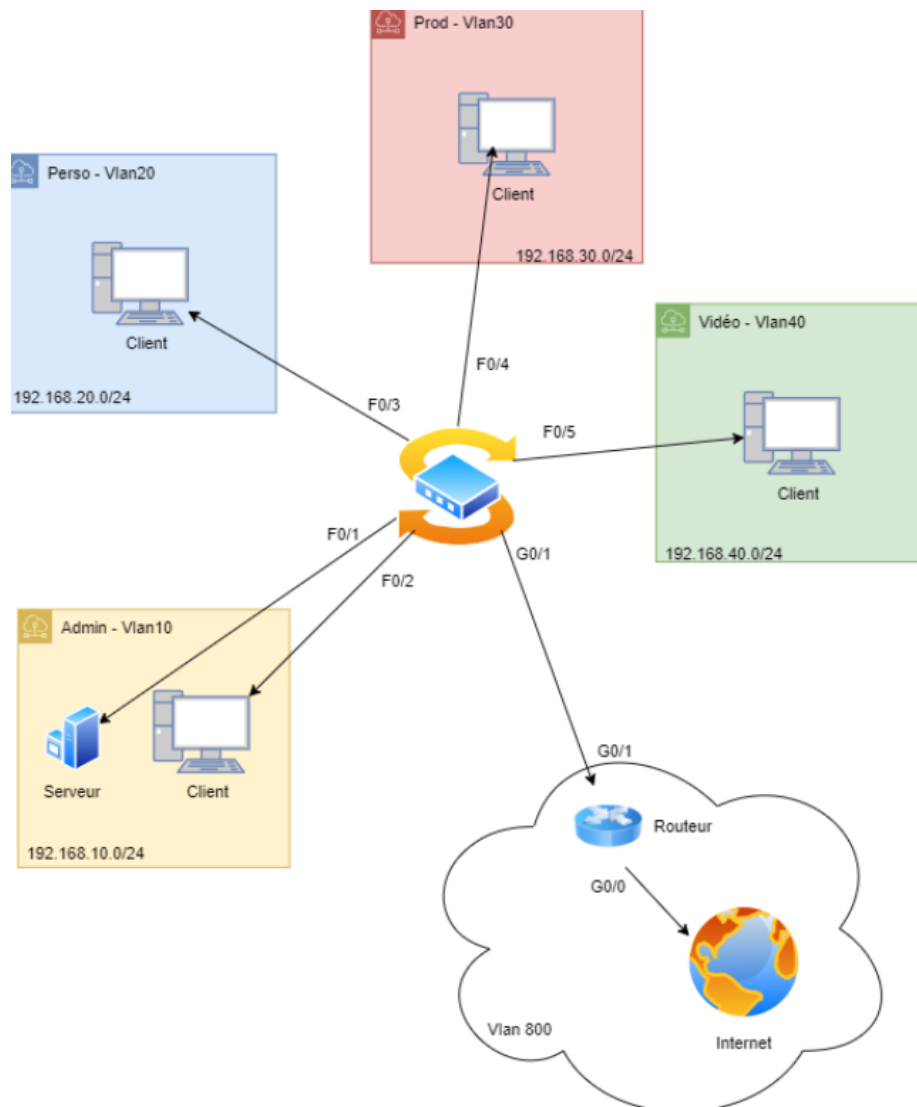
Recherches Matérielles

Voir les PDF:

- RJ-45
- Routeur Cisco 2900
- Switch Cisco 2960

du Livrable 1

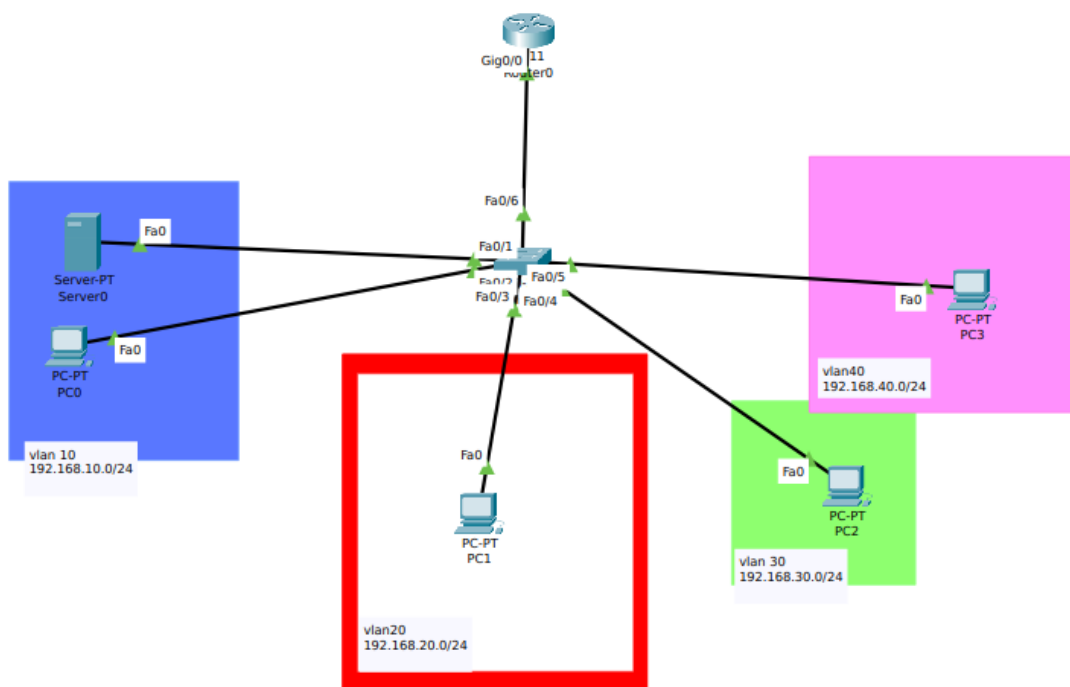
Draw.io



4 réseaux différents :

- ADMIN
- PERSONNEL : permet aux machines des employés de dialoguer entre elles
- PRODUCTION
- VIDEO

Maquette Cisco



Configuration VLAN, DNS, DHCP, ACL (cisco)

Routeur:

DHCP:

```
ip dhcp pool vlan10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 10.0.0.1
ip dhcp pool vlan20
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
ip dhcp pool vlan30
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
ip dhcp pool vlan40
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
*****

interface GigabitEthernet0/0.1
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
ip access-group VLAN10_ACL in
!
interface GigabitEthernet0/0.2
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
ip access-group VLAN20_ACL in
!
interface GigabitEthernet0/0.3
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
ip access-group VLAN30_ACL in
!
interface GigabitEthernet0/0.4
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
ip access-group VLAN40_ACL in
*****
```

ACL:

```
ip access-list extended VLAN20_ACL
deny icmp any any echo
permit ip any any
```

```
ip access-list extended VLAN30_ACL
deny icmp any any echo
permit ip any any
ip access-list extended VLAN40_ACL
deny icmp any any echo
permit ip any any
ip access-list extended VLAN10_ACL
permit icmp any any echo
permit ip any any
```

Switch:

```
interface FastEthernet0/3
switchport access vlan 10
switchport mode access
```

```
interface FastEthernet0/1
switchport access vlan 20
switchport mode access
```

```
interface FastEthernet0/2
switchport access vlan 30
switchport mode access
```

```
interface FastEthernet0/4
switchport access vlan 40
switchport mode access
interface FastEthernet0/5
switchport mode trunk
```

Ping capture de trames (cisco)

```
C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Control-C
^C
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.30.1: bytes=32 time<1ms TTL=255
Reply from 192.168.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.30.1:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Control-C
^C
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Test VM

```
administrateur@rt-mv:~$ ping 192.168.40.2
PING 192.168.40.2 (192.168.40.2) 56(84) bytes of data.
64 bytes from 192.168.40.2: icmp_seq=1 ttl=63 time=0.862 ms
64 bytes from 192.168.40.2: icmp_seq=2 ttl=63 time=0.859 ms
64 bytes from 192.168.40.2: icmp_seq=3 ttl=63 time=0.966 ms
64 bytes from 192.168.40.2: icmp_seq=4 ttl=63 time=0.919 ms
^C
--- 192.168.40.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.859/0.901/0.966/0.044 ms
administrateur@rt-mv:~$ ping 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=63 time=1.49 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=63 time=1.32 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=63 time=0.870 ms
^C
--- 192.168.20.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2017ms
rtt min/avg/max/mdev = 0.870/1.226/1.488/0.261 ms
administrateur@rt-mv:~$ ping 192.168.30.2
PING 192.168.30.2 (192.168.30.2) 56(84) bytes of data.
64 bytes from 192.168.30.2: icmp_seq=1 ttl=63 time=1.21 ms
64 bytes from 192.168.30.2: icmp_seq=2 ttl=63 time=1.20 ms
64 bytes from 192.168.30.2: icmp_seq=3 ttl=63 time=1.24 ms
64 bytes from 192.168.30.2: icmp_seq=4 ttl=63 time=1.39 ms
64 bytes from 192.168.30.2: icmp_seq=5 ttl=63 time=1.19 ms
^C
--- 192.168.30.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4073ms
rtt min/avg/max/mdev = 1.191/1.246/1.392/0.075 ms
```

vlan10 → vlan20, 30, 40

```

root@rt-mv:/home/administrateur# ping 192.168.30.2
PING 192.168.30.2 (192.168.30.2) 56(84) bytes of data.
From 192.168.40.1 icmp_seq=1 Packet filtered
From 192.168.40.1 icmp_seq=2 Packet filtered
From 192.168.40.1 icmp_seq=3 Packet filtered
^C
--- 192.168.30.2 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2075ms

Wireshark :/home/administrateur# ping 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
From 192.168.40.1 icmp_seq=1 Packet filtered
From 192.168.40.1 icmp_seq=2 Packet filtered
From 192.168.40.1 icmp_seq=3 Packet filtered
^C
--- 192.168.20.2 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2059ms

root@rt-mv:/home/administrateur# ping 192.168.10.2
PING 192.168.10.2 (192.168.10.2) 56(84) bytes of data.
From 192.168.40.1 icmp_seq=1 Packet filtered
From 192.168.40.1 icmp_seq=2 Packet filtered
From 192.168.40.1 icmp_seq=3 Packet filtered
^C
--- 192.168.10.2 ping statistics ---
3 packets transmitted, 0 received, +3 errors, 100% packet loss, time 2035ms

root@rt-mv:/home/administrateur#

```

vlan40 → vlan10, 20, 30

Serveur Web Apache (VM)

Installation du serveur Apache :

sudo apt update

sudo apt install apache2

sudo systemctl status apache2

```

administrateur@rt-mv:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2023-12-19 14:25:58 CET; 30s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3221 (apache2)
    Tasks: 55 (limit: 2878)
   Memory: 5.0M
      CPU: 21ms
   CGroup: /system.slice/apache2.service
           └─3221 /usr/sbin/apache2 -k start
             └─3222 /usr/sbin/apache2 -k start
               └─3223 /usr/sbin/apache2 -k start

```

```
sudo ufw allow 'Apache'
```

```
sudo ufw allow 'Apache'
```

Pour la modification du serveur Apache :

```
cd /var/www/html
```

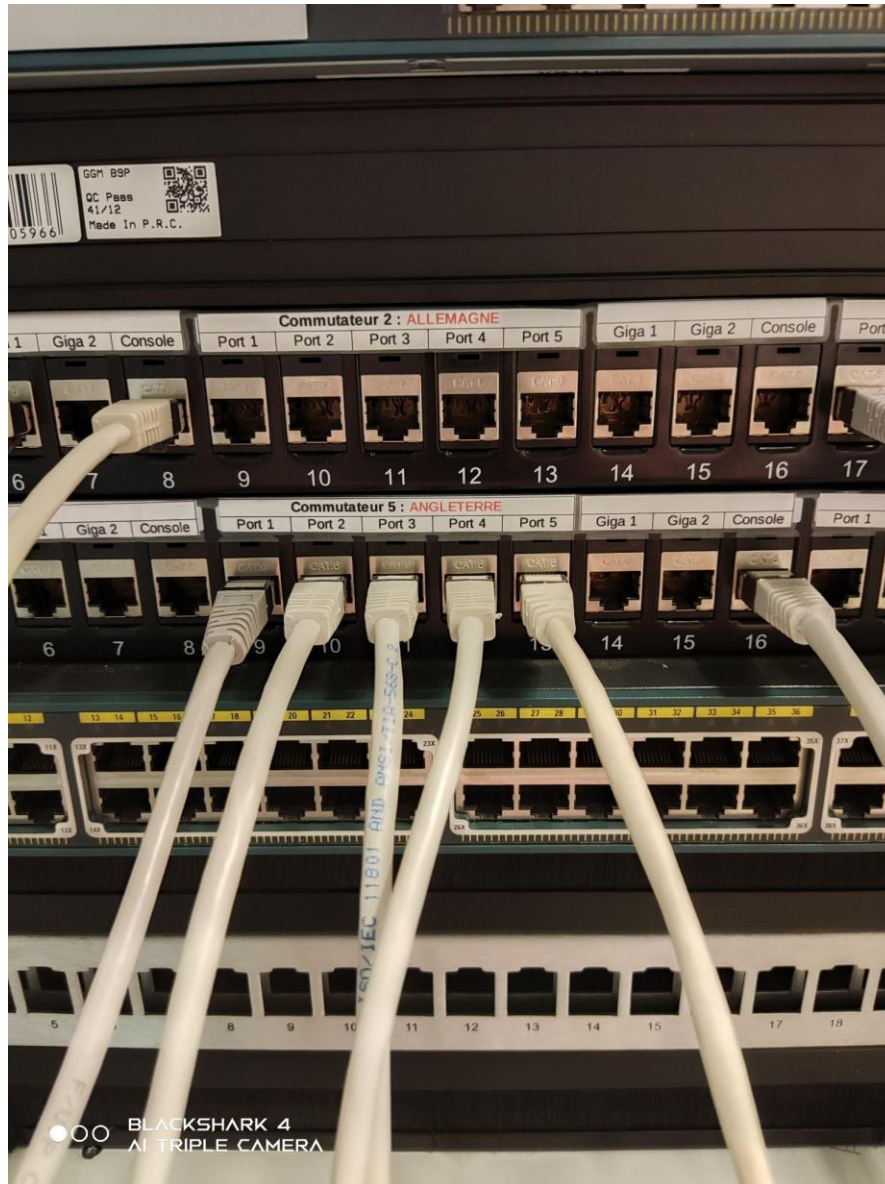
```
sudo -s
```

nano index.html

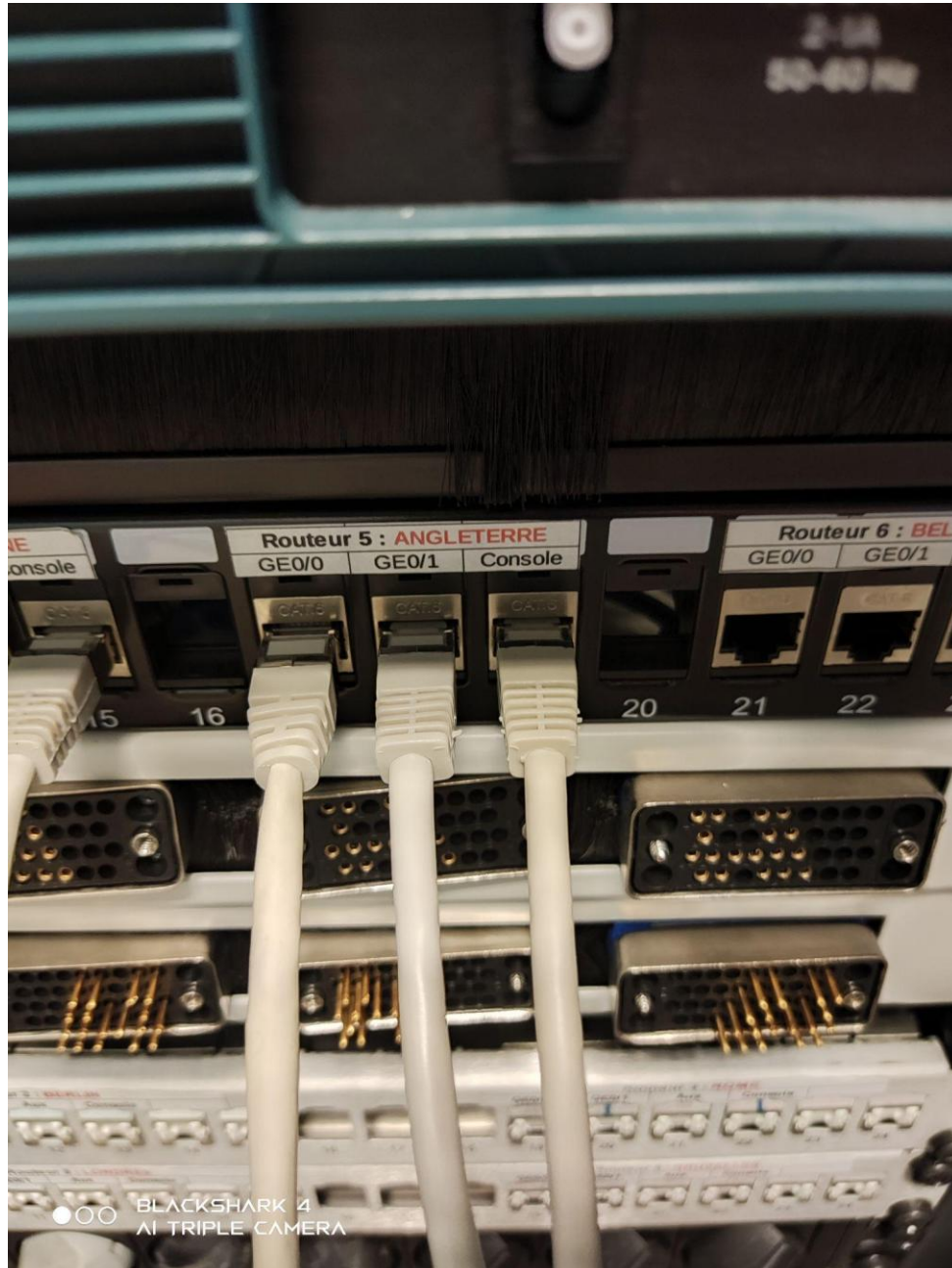
```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

```

Câblage (réel)



Câblage sur le switch



Câblage routeur

Configuration VLAN, DHCP, DNS, ACL, NAT (réel)

Switch :

```
interface FastEthernet0/3
switchport access vlan 10
switchport mode access
interface FastEthernet0/1
switchport access vlan 20
switchport mode access
interface FastEthernet0/2
switchport access vlan 30
switchport mode access
interface FastEthernet0/4
switchport access vlan 40
switchport mode access
interface FastEthernet0/5
switchport mode trunk
```

Routeur:

```
p dhcp excluded-address 192.168.10.1
ip dhcp excluded-address 192.168.20.1
ip dhcp excluded-address 192.168.30.1
ip dhcp excluded-address 192.168.40.1
```

```
ip dhcp pool vlan10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 10.0.0.1
ip dhcp pool vlan20
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 10.0.0.1
ip dhcp pool vlan30
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
```

```
dns-server 10.0.0.1
ip dhcp pool vlan40
network 192.168.40.0 255.255.255.0
default-router 192.168.40.1
dns-server 10.0.0.1
```

```
interface GigabitEthernet0/0.1
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
 ip access-group VLAN10_ACL in
 ip nat inside
 ip virtual-reassembly in
interface GigabitEthernet0/0.2
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
 ip access-group VLAN20_ACL in
 ip nat inside
 ip virtual-reassembly in
interface GigabitEthernet0/0.3
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
 ip access-group VLAN30_ACL in
 ip nat inside
 ip virtual-reassembly in
interface GigabitEthernet0/0.4
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
 ip access-group VLAN40_ACL in
 ip nat inside
 ip virtual-reassembly in
interface GigabitEthernet0/1
 ip address dhcp
 ip nat outside
 ip virtual-reassembly in
duplex auto
speed auto
```

```
ip nat inside source list 1 interface GigabitEthernet0/1 overload
ip nat inside source list 2 interface GigabitEthernet0/1 overload
ip nat inside source list 3 interface GigabitEthernet0/1 overload
ip nat inside source list 4 interface GigabitEthernet0/1 overload
```

```
ip access-list extended VLAN10_ACL
  permit icmp any any echo
  permit ip any any
ip access-list extended VLAN20_ACL
  deny  icmp any any echo
  permit ip any any
ip access-list extended VLAN30_ACL
  deny  icmp any any echo
  permit ip any any
ip access-list extended VLAN40_ACL
  deny  icmp any any echo
  permit ip any any
access-list 1 permit 192.168.10.0 0.0.0.255
access-list 2 permit 192.168.20.0 0.0.0.255
access-list 3 permit 192.168.30.0 0.0.0.255
access-list 4 permit 192.168.40.0 0.0.0.255
```


Serveur Web Apache (réel)

Installation du serveur Apache :

```
sudo apt update
```

```
sudo apt install apache2
```

```
sudo systemctl status apache2
```

```

administrateur@rt-mv:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2023-12-19 14:25:58 CET; 30s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 3221 (apache2)
    Tasks: 55 (limit: 2878)
   Memory: 5.0M
      CPU: 21ms
   CGroup: /system.slice/apache2.service
           └─3221 /usr/sbin/apache2 -k start
             └─3222 /usr/sbin/apache2 -k start
               └─3223 /usr/sbin/apache2 -k start

```

```
sudo ufw allow 'Apache'
```

```
administrateur@rt-mv:~$ sudo ufw allow 'Apache'
Les règles ont été mises à jour
Les règles ont été mises à jour (IPv6)
```

Pour la modification du serveur Apache :

```
cd /var/www/html
```

```
sudo -s
```

nano index.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/
<html xmlns="http://www.w3.org/1999/xhtml">
<!--
  Modified from the Debian original for Ubuntu
  Last updated: 2022-03-22
  See: https://launchpad.net/bugs/1966004
-->
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
  <title>Apache2 Ubuntu Default Page: It works</title>
  <style type="text/css" media="screen">
* {
  margin: 0px 0px 0px 0px;
  padding: 0px 0px 0px 0px;
}
```

Ping, capture de trames (réel)

24	9.091638131	192.168.10.3	192.168.40.2	ICMP	98 Echo (ping) request	id=0x0002, seq=1/256, ttl=64 (reply in 25)
25	9.092969894	192.168.40.2	192.168.10.3	ICMP	98 Echo (ping) reply	id=0x0002, seq=1/256, ttl=63 (request in 24)
31	10.104816137	192.168.10.3	192.168.40.2	ICMP	98 Echo (ping) request	id=0x0002, seq=2/512, ttl=64 (reply in 32)
32	10.105585943	192.168.40.2	192.168.10.3	ICMP	98 Echo (ping) reply	id=0x0002, seq=2/512, ttl=63 (request in 31)
33	11.159997503	192.168.10.3	192.168.40.2	ICMP	98 Echo (ping) request	id=0x0002, seq=3/768, ttl=64 (reply in 34)
34	11.160781199	192.168.40.2	192.168.10.3	ICMP	98 Echo (ping) reply	id=0x0002, seq=3/768, ttl=63 (request in 33)
38	12.199046423	192.168.10.3	192.168.40.2	ICMP	98 Echo (ping) request	id=0x0002, seq=4/1024, ttl=64 (reply in 39)
39	12.199874074	192.168.40.2	192.168.10.3	ICMP	98 Echo (ping) reply	id=0x0002, seq=4/1024, ttl=63 (request in 38)